

Tau-5* structural ensembles

Chemical shifts, SAXS and cooperative helicity

Methods • equations • theory • current results

WT and W397A/W433A (AA) | 29 September 2026

Zhu et al., Nature Communications (2022) • Current analysis: code, selected members and numerical results

The structural question

Does SAXS compaction arise from long cooperative helices and hydrophobic contacts, or from collapsed disordered tails?

Chemical shifts constrain local environments; SAXS constrains global size and pair-distance distributions.

Different ensembles can satisfy the same averages. Pool design is therefore part of the inference.

Constructs and residue mapping

Region / construct	AR numbering	Model numbering
Full Tau-5*	GP + 330–447	1–120
AA substitutions	W397A / W433A	70 / 106
Matched R2_R3 segment	391–446 (56 residues)	64–119
CS comparison range	330–446	3–119

Use each construct's own sequence and available shifts; omit terminal predictions.

Current analysis: code, selected members and numerical results · Zhu et al., Nature Communications (2022)

Current workflow

- 1 Generate 2,000 IDPConfGen conformers per construct; add the helix-length supplement for an expanded comparison.
- 2 Predict shifts and SAXS for every conformer; average observables over each candidate subset.
- 3 Select 500 distinct conformers, then inspect shifts, SAXS, helicity, size and contacts.

Ensemble averages and identifiability

$$\langle \mathbf{A}_j \rangle = \int \mathbf{A}_j(\mathbf{x}) p(\mathbf{x}) d\mathbf{x} \approx \sum_k w_k \mathbf{A}_j(\mathbf{x}_k)$$

$\mathbf{x}_k$: conformer; $\mathbf{A}_j$: predicted observable; $w_k \geq 0$ and $\sum_k w_k = 1$.

An observable constrains an average, usually not a unique structural distribution.

A finite pool limits reachable averages to its convex hull. Equal-weight subsets impose additional restrictions.

IDPConfGen proposal ensembles

Statistical backbone fragments generate diverse conformations for each exact sequence.

Residue-specific $\delta 2D$ populations guide the secondary-structure proposal probabilities.

The output must be checked with DSSP and predicted shifts: proposal probabilities need not equal realized helix populations.

δ 2D populations and realized helicity

$$h_r = (1/N) \sum_k \mathbb{1}[\text{DSSP}_k(r) = H]$$

h_r : observed α -helix fraction at residue r ; N : number of structures.

δ 2D infers secondary-structure populations from chemical shifts; DSSP labels hydrogen-bond geometry in structures.

Agreement is a useful diagnostic, but these two definitions are not identical observables.

Helix-length enrichment: current implementation

Parameter	Current setting
Eligible centers	$\delta 2D$ helix population ≥ 0.10
Imposed core lengths	4, 6, 8, 10, 12 residues
Core torsions	$\phi = -63^\circ \pm 5^\circ$; $\psi = -42^\circ \pm 5^\circ$; $\omega = 180^\circ$
Supplement size	WT +110; AA +90 conformers

These are geometric supplements; equal counts per window do not preserve the target profile.

Current analysis: code, selected members and numerical results

Helix centers are blurred into residue occupancy

$$h_r \approx (1/N) \sum_{c,l} n_{c,l} \mathbb{1}[r \in \text{window}(c,L)]$$

$n_{c,l}$: count of helices centered at c with length L ; N : pool size.

Long helices contribute to neighboring residues. Sampling centers in proportion to δ^2D alone broadens the profile.

Choose center/length counts using the full coverage matrix, then verify actual DSSP populations.

Current analysis: code, selected members and numerical results

Post-hoc pool calibration against $\delta 2D$

$$\min \Sigma_r \mid N^{-1} \Sigma_k z_k H_{k_r} - h_r^{D2D} \mid$$

$z_k \in \{0,1\}$; $\sum_k z_k = N$. H_{k_r} is the measured DSSP-H indicator.

Select structures to approximate a target residue profile while retaining equal weights.

A separate calibration pilot exists; it is not the input to the reported base/expanded CS-SAXS selections.

Current analysis: code, selected members and numerical results

Secondary chemical shifts

$$\Delta\delta_{r,a} = \delta_{r,a} - \delta_{r,a}^{Rc}$$

r: residue; a: nucleus; δ^{Rc} : sequence-specific SPARTA+ random-coil reference.

Use the same baseline for experiment and calculation within each construct.

$\Delta\delta(\text{exp}) - \Delta\delta(\text{calc}) = \delta(\text{exp}) - \delta(\text{calc})$: subtraction clarifies plots without changing residuals.

Prediction error for an ensemble average

$$\text{Var}(\bar{\epsilon}) = N^{-2} \sum_k \sum_l \text{Cov}(\epsilon_k, \epsilon_l)$$

ϵ_k : prediction error for conformer k. Independent errors average down; shared systematic errors remain.

Predictor RMSE on folded structures is not automatically the uncertainty of an IDP ensemble mean.

Inspect coherent residue-wise errors and multiple nuclei; use secondary shifts to remove distracting chemical-identity baselines.

SAXS measures pair-distance information

$$I(q) = \sum_i \sum_j f_i(q) f_j(q) \sin(qr_{ij})/(qr_{ij})$$

q : scattering-vector magnitude; r_{ij} : atom separation; f_i : atomic scattering factor.

The Debye expression explains structural sensitivity; CRY SOL also treats solvent exclusion and hydration contrast.

For an ensemble: $\bar{I}(q) = N^{-1} \sum_k I_k(q)$. Average intensities before fitting the scale.

CRY SOL back-calculation

Setting	Value
Angular / Fibonacci orders	lm = 20; fb = 17
q grid	501 points; qmax = 0.5 Å ⁻¹
Solvent / hydration density contrast	0.334 / 0.03 e Å ⁻³
Fit nuisance parameters	One nonnegative scale; background fixed to zero

Per-conformer hydration optimization is disabled.

Svergun et al., J. Appl. Crystallogr. (1995) · Current analysis: code, selected members and numerical results

Joint CS–SAXS selection objective

$$J(S) = \sum_j [(\delta_j(S) - \delta_j^{\text{exp}})/s_{a(j)}]^2 + \sum_m [(a \bar{I}_m(S) - I_m^{\text{exp}})/\sigma_m]^2$$

S: 500-member subset; j: available shift; m: SAXS point.

s_a : chosen shift scale for nucleus type a; σ_m : experimental SAXS error.

The current objective sums residuals directly; it does not normalize each data type by its number of points.

Current analysis: code, selected members and numerical results · Schwalbe et al., Structure (2014)

Analytical SAXS scale factor

$$a^* = \max[0, (\sum_m \bar{I}_m I_m^{\text{exp}} / \sigma_m^2) / (\sum_m \bar{I}_m^2 / \sigma_m^2)]$$

The scale minimizes the weighted squared SAXS residual for a fixed subset.

$\bar{I}_m$: ensemble-mean calculated intensity; I_m^{exp} : experiment; σ_m : experimental uncertainty.

Reported $\chi^2/\text{point} = (1/M) \sum_m [(a^* \bar{I}_m - I_m^{\text{exp}}) / \sigma_m]^2$. It is not a degrees-of-freedom-corrected reduced χ^2 .

Current analysis: code, selected members and numerical results

Relative weights in the current fit

Data type	Scale	Role
C α , C', C β	0.10 ppm	Strong carbon-shift constraint
N	0.50 ppm	Nitrogen-shift constraint
HN, H α	0.20 ppm	Proton constraint when available
SAXS	Measured $\sigma(q)$	Point-wise scattering residual

Contribution $\propto$ number of observations $\times$ (RMSD / scale)².

Current analysis: code, selected members and numerical results · Cavender et al., LiveCoMS (2025)

Equal-weight genetic selection

$$w_k = 1/500 \text{ for } k \in S; \quad w_k = 0 \text{ otherwise}$$

Each candidate contains 500 distinct conformers. Selection changes membership, not individual weights.

Population 100; 1,000 generations; mutations replace 1, 2, 5 or 25 members; crossover samples parental unions.

Three optimizer seeds are run on the same candidate pool for each construct and pool type.

Ozenne et al., JACS (2012) · Schwalbe et al., Structure (2014) · Current analysis: code, selected members and numerical results

Relationship to ASTEROIDS and EOM

ASTEROIDS: select ensembles against experimental averages, using statistical conformer pools and multiple structural observables.

EOM: select ensembles primarily from SAXS agreement; inspect the selected size and shape distributions.

Here: an ASTEROIDS-style equal-weight subset search with explicit CS and SAXS residuals. It is not the original ASTEROIDS software.

Continuous reweighting: comparison framework

$$\min \frac{1}{2}\chi^2(\mathbf{w}) + \theta \sum_k w_k \ln(w_k / w_k^0)$$

$w_k \geq 0$; $\sum_k w_k = 1$. θ sets the penalty for departing from prior weights w_k^0 .

$N_{\text{eff}} = 1 / \sum_k w_k^2$ measures weight concentration; an equal-weight 500-member subset has $N_{\text{eff}} = 500$.

This is a generic regularized reweighting formulation. The current equal-weight search has no such entropy term.

Radius of gyration and SAXS size

$$R_g = [N \alpha^{-1} \sum_i |r_i - \bar{r}|^2]^{1/2}$$

r_i : Ca coordinate; $\bar{r}$: Ca centroid; N : number of selected residues.

The reported structural R_g uses equally weighted Ca atoms and is expressed in Å.

Guinier theory: $I(q) \approx I(0) \exp(-q^2 R_g^2/3)$ at small q . Scattering-weighted R_g need not equal the Ca geometric value.

S α : the exact AR analysis definition

$$S\alpha = \sum_u [1 - (R_u/r_0)^8] / [1 - (R_u/r_0)^{12}]$$

u : seven-residue window; R_u : aligned backbone RMSD to a helical reference; $r_0 = 0.10$ nm.

Use N, C α , C and O atoms (28/window), with Kabsch alignment. Full chain: 114 windows; R2_R3: 50.

Each window contributes 1 at perfect agreement and 2/3 at $R_u = r_0$. $S\alpha$ is not a residue helix fraction.

Original AR analysis code and MD comparison · Zhu et al., Nature Communications (2022)

Contact maps: the exact AR analysis definition

$$C_{i,j} = (1/N) \sum_k \mathbb{1}[\min_{a \in i, b \in j} d_{ab}(x_k) < 1.2 \text{ nm}]$$

i, j : residues; a, b : heavy atoms; N : conformer count. Diagonal entries are set to zero.

Keep adjacent residue pairs. Compare full chains and the identical AR 391–446 segment separately.

A 1.2 nm cutoff is a broad proximity measure; it does not isolate aromatic stacking or a specific interaction energy.

Original AR analysis code and MD comparison · Zhu et al., Nature Communications (2022)

Current selection experiment

Construct	Base pool	Expanded pool	Selected
WT	2,000	2,110	500
W397A/W433A	2,000	2,090	500

2 sequences × 2 pool definitions × 3 optimizer seeds = 12 selections

All seeds reuse the same candidate pool within a condition.

Current analysis: code, selected members and numerical results

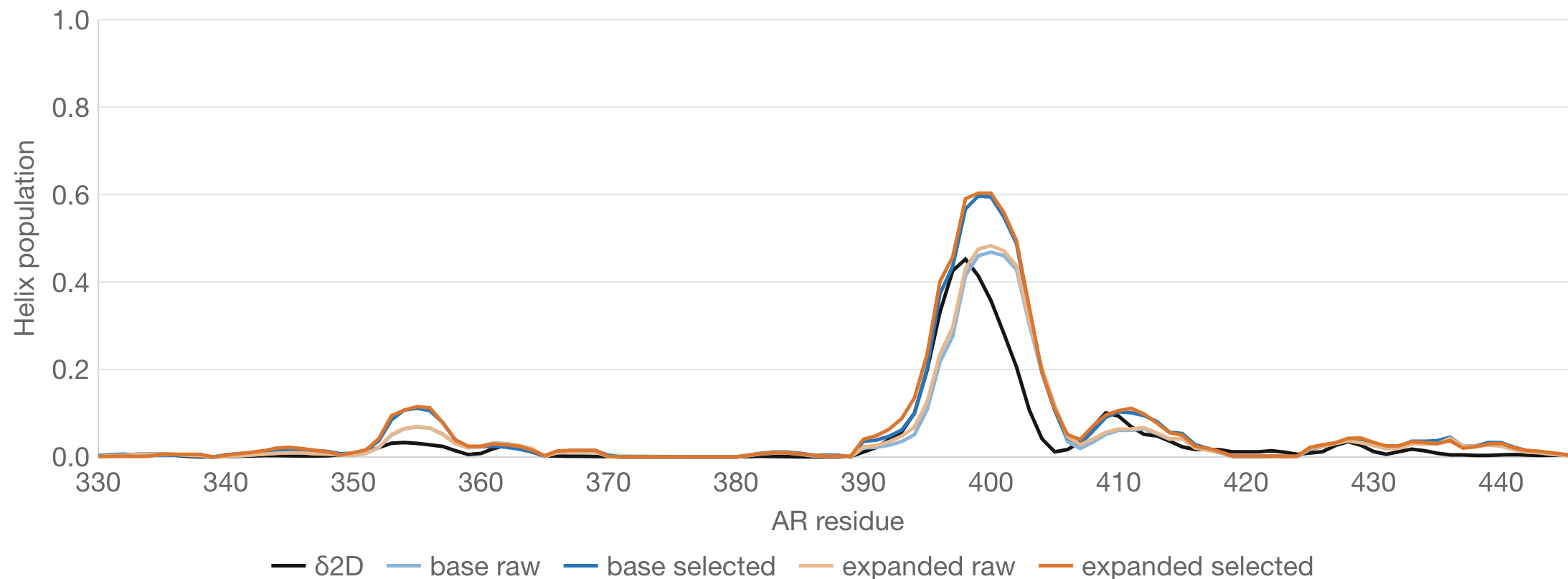
Carbon fits improve; WT SAXS remains limited

Condition	C' RMSD	C α RMSD	C β RMSD	SAXS χ^2 /pt
WT base	0.420–0.428	0.300–0.307	—	7.74–7.94
WT expanded	0.416–0.420	0.296–0.303	—	7.34–7.90
AA base	0.396–0.403	0.310–0.315	0.287–0.295	2.85–2.95
AA expanded	0.395–0.399	0.307–0.310	0.290–0.293	2.77–2.86

The current supplement gives modest changes, not a decisive improvement.

Current analysis: code, selected members and numerical results

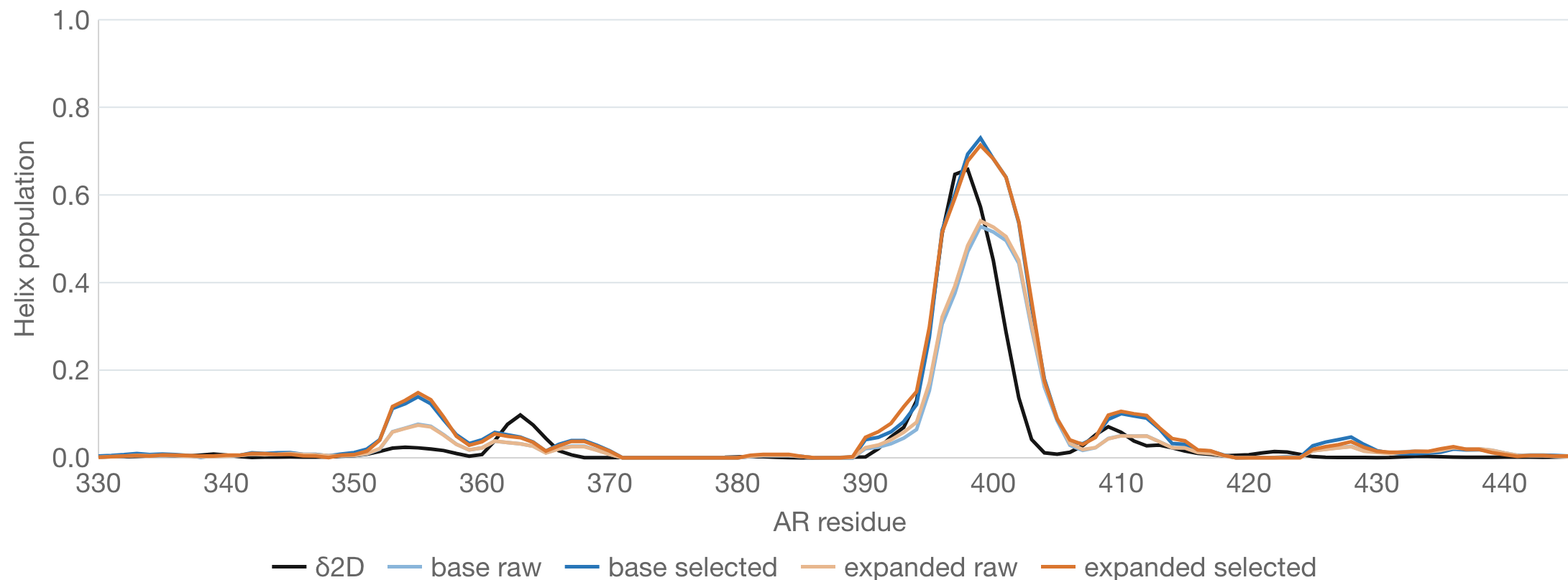
WT: whole-sequence helix populations



DSSP α -helix (H); selection curves averaged across three optimizer seeds.

Current analysis: code, selected members and numerical results · Camilloni et al., Biochemistry (2012)

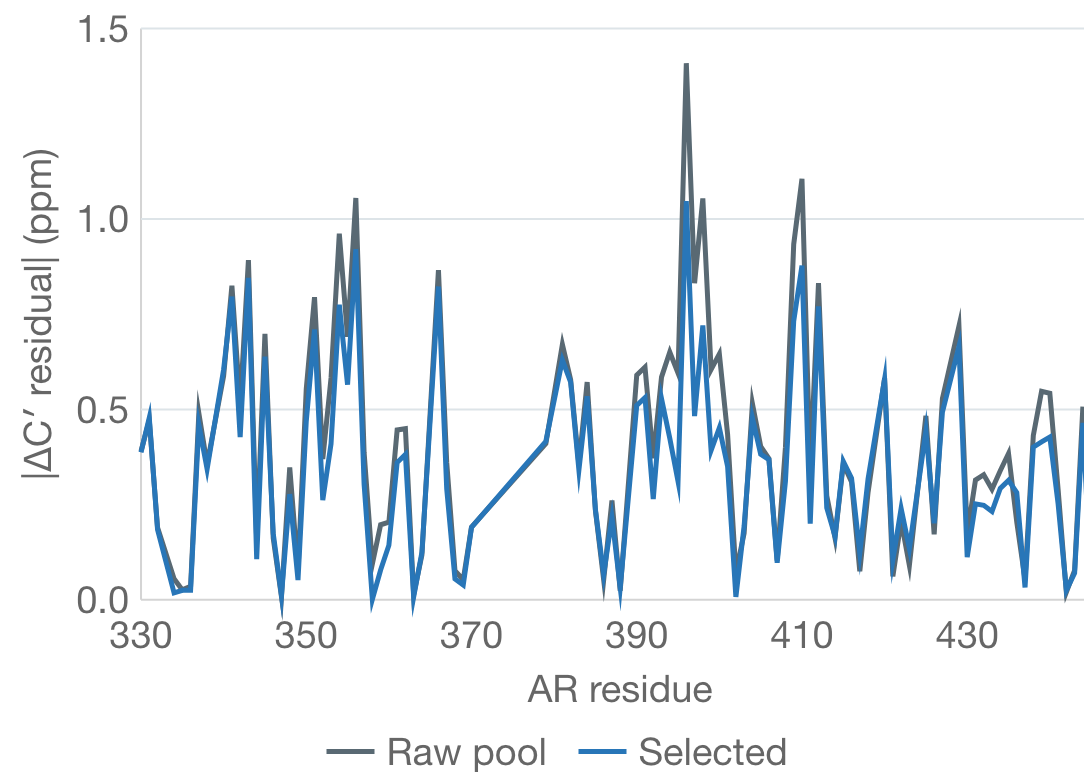
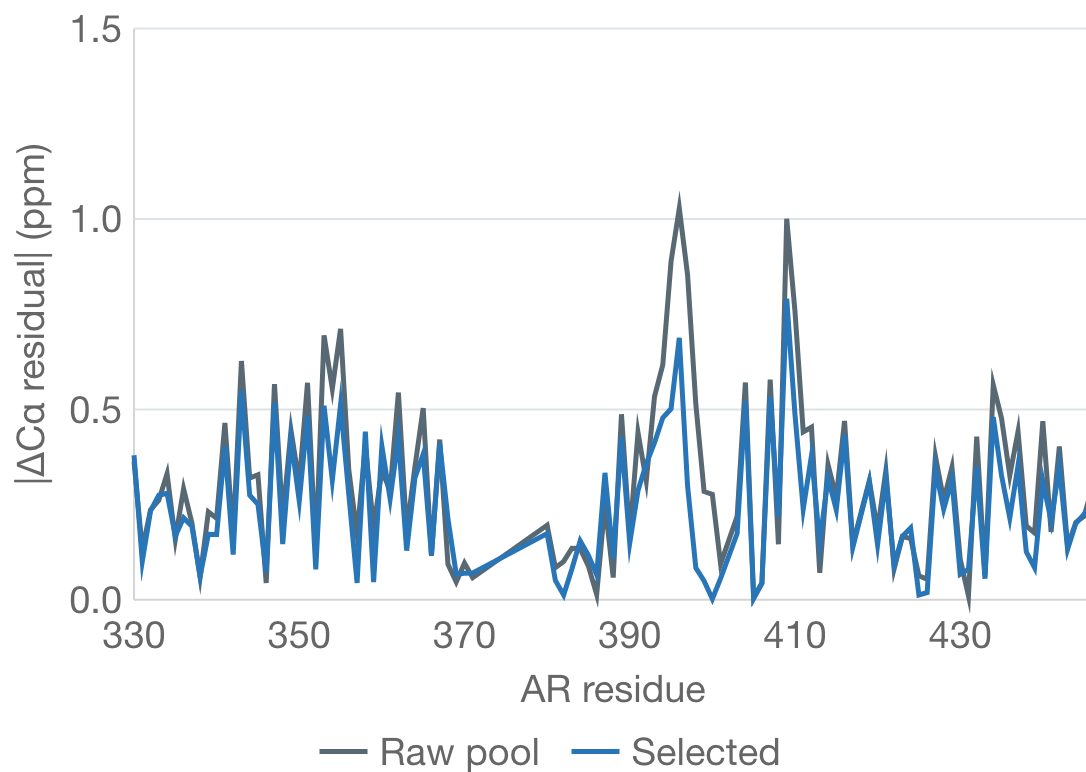
AA: whole-sequence helix populations



DSSP α -helix (H); selection curves averaged across three optimizer seeds.

Current analysis: code, selected members and numerical results · Camilloni et al., Biochemistry (2012)

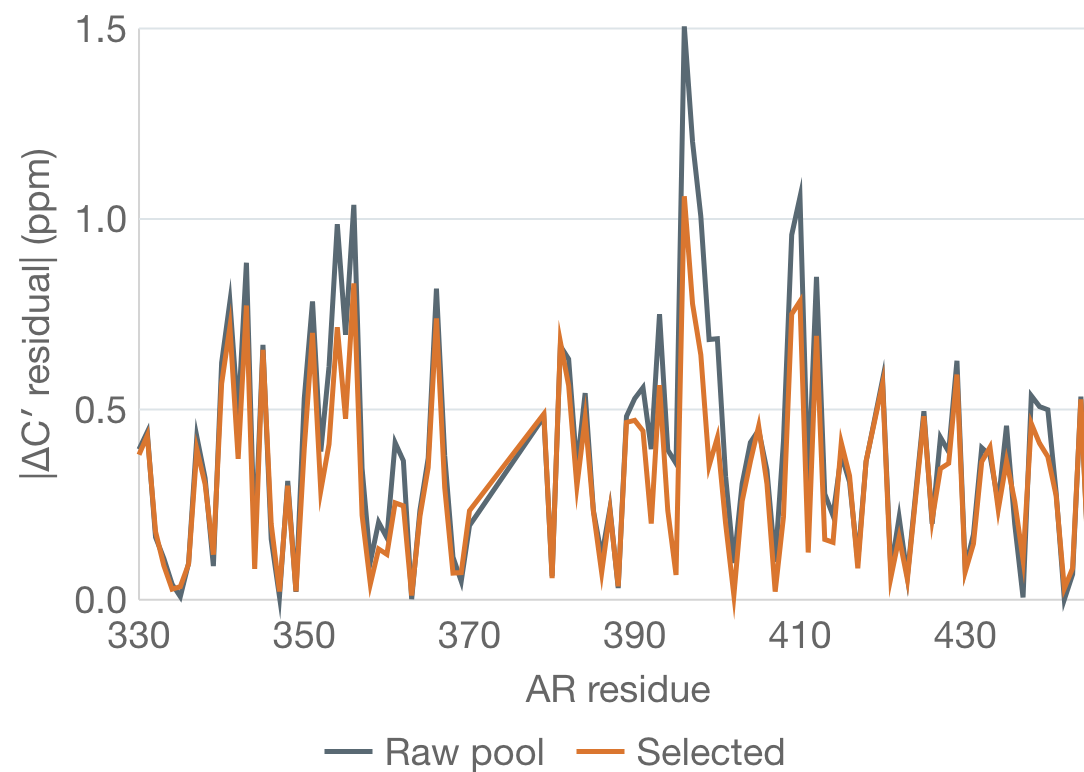
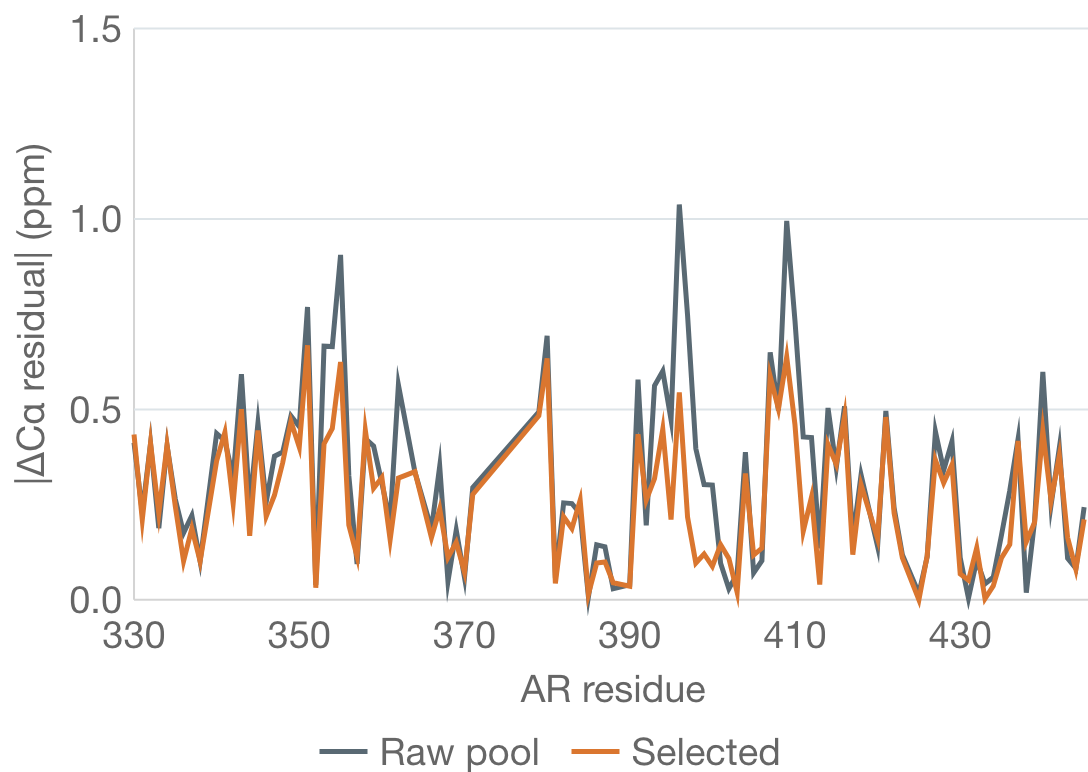
WT: carbon absolute residuals



Expanded pool, seed 20260929. Fitting averages can retain localized errors.

Current analysis: code, selected members and numerical results

AA: carbon absolute residuals



Expanded pool, seed 20260929. Fitting averages can retain localized errors.

Current analysis: code, selected members and numerical results

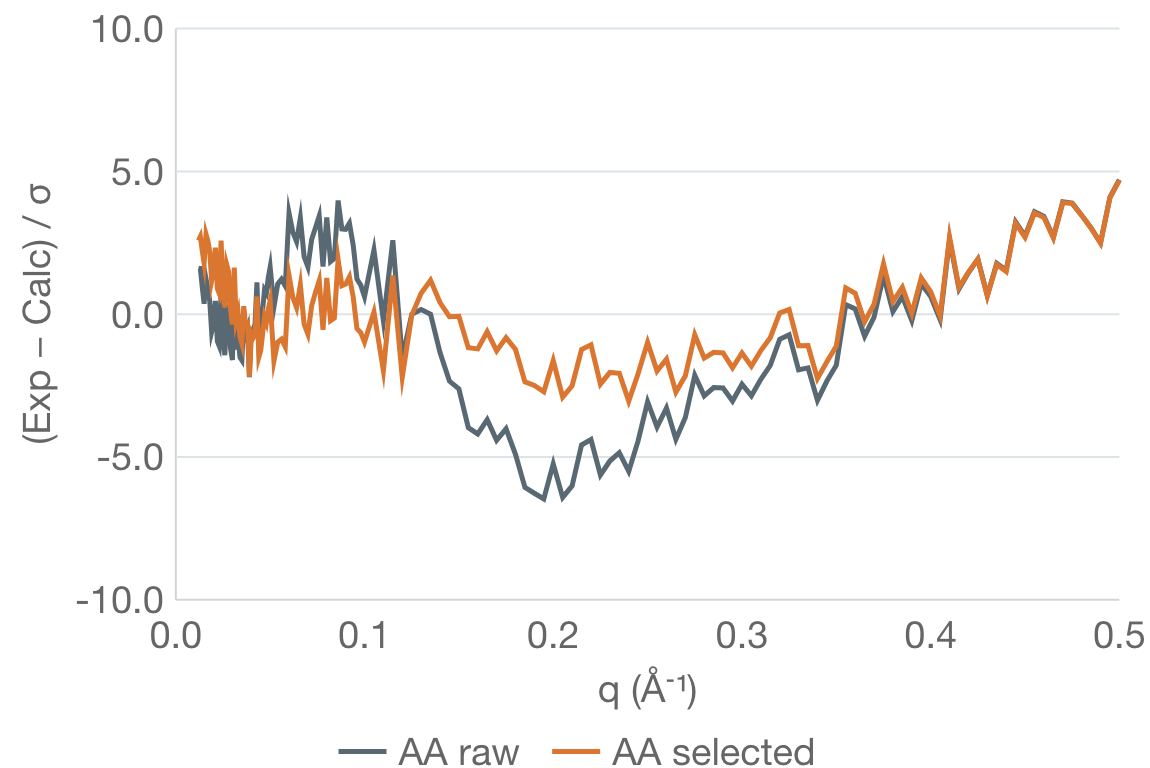
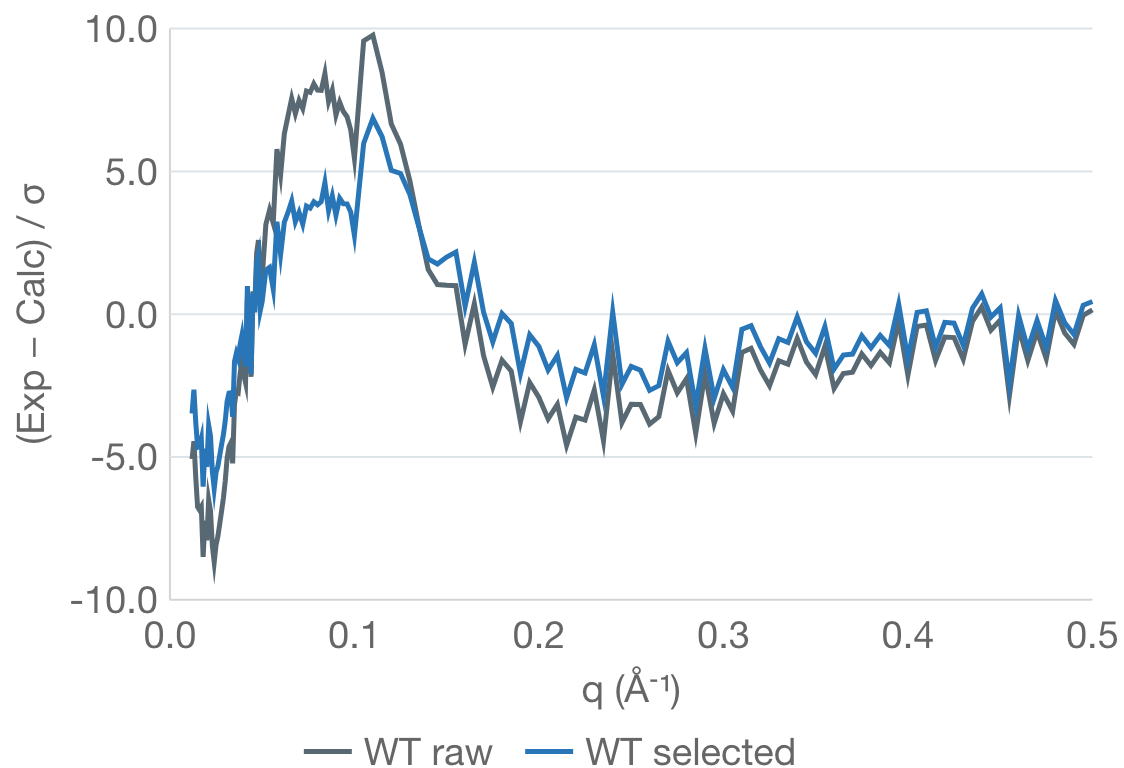
Remaining N and HN residuals

Condition	N RMSD (ppm)	HN RMSD (ppm)
WT base	0.779–0.792	0.217–0.219
WT expanded	0.763–0.778	0.217–0.218
AA base	1.324–1.351	0.155–0.158
AA expanded	1.321–1.338	0.156–0.158

AA nitrogen remains the largest nucleus-specific residual in ppm.

Current analysis: code, selected members and numerical results

SAXS: residual structure persists



Structured residuals indicate that mean size alone is not a sufficient SAXS check.

Current analysis: code, selected members and numerical results

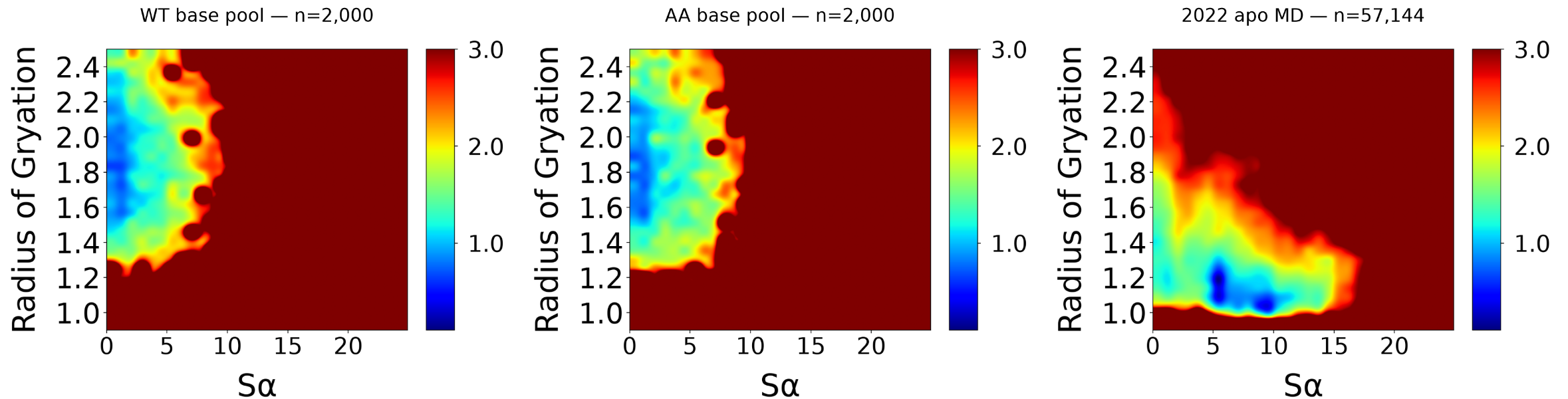
Selection repeatability and convergence

Condition	Max helix seed spread	Mean C α Rg across seeds
WT base	0.040	31.99–32.20 Å
WT expanded	0.036	32.00–32.06 Å
AA base	0.022	31.72–31.77 Å
AA expanded	0.026	31.64–31.76 Å

Objective still improves in the last 100 generations. No independent-pool convergence claim.

Current analysis: code, selected members and numerical results

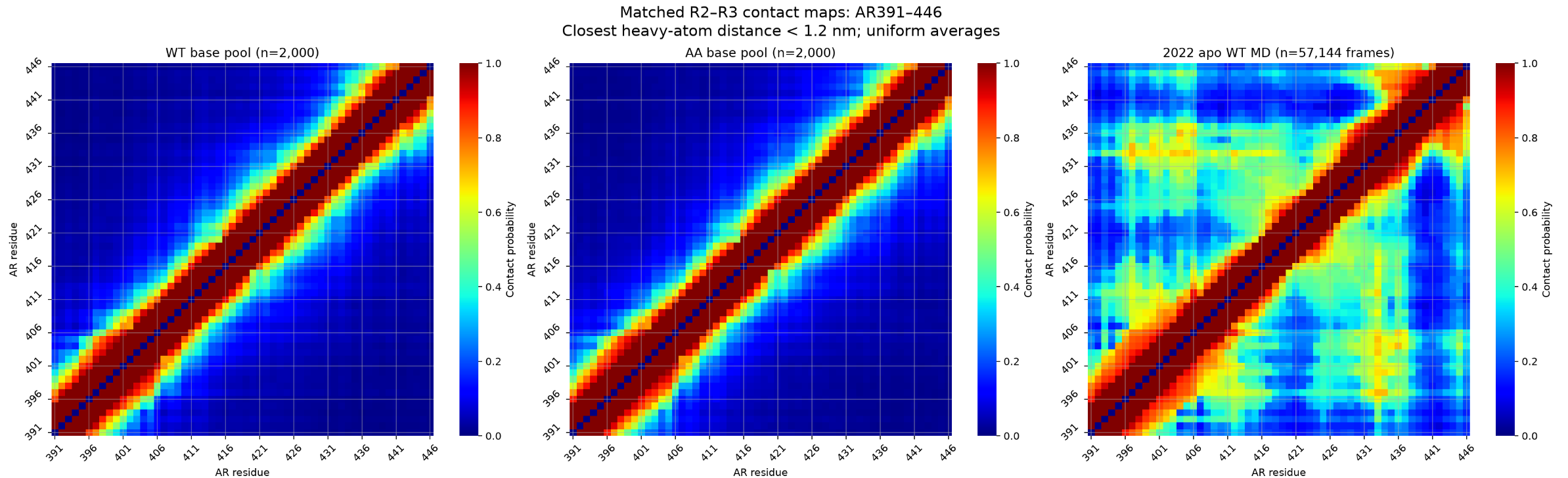
Raw pools versus the published MD ensemble



AR 391–446; Rg in nm. Color: $-RT \ln(\text{density} + 10^{-6})$, $T = 300$ K. For raw pools this is a display transform, not a thermodynamic free energy.

Original AR analysis code and MD comparison · Zhu et al., Nature Communications (2022)

R2_R3 contact maps



Identical residue segment and contact definition. MD is apo WT; no AA MD comparison is implied.

Original AR analysis code and MD comparison · Zhu et al., Nature Communications (2022)

bAles: structural prior

$$p_0(\mathbf{x}) \propto \exp[-E^{Rc}(\mathbf{x})/(kBT)]$$
$$E^{Rc} = E^{ff} + \sum_i E^{cMAP}(\phi_i, \psi_i)$$

The random-coil prior combines bonded terms, repulsive nonbonded interactions and residue-dependent torsion corrections.

$E^{cMAP} = -kBT \ln(P_{coil} / P_{FF})$ adjusts torsion sampling toward coil statistics.

This supplies a physical sampling model for the AF2-informed posterior.

bAles: distogram likelihood

$$L_i(x) = \text{erf}[\Delta_i / (\sqrt{2} \sigma_{i,\text{min}})] / (2\Delta_i)$$
$$E_{\text{post}} = E^{\text{Rc}} - kBT \sum_i \ln L_i + \text{constant}$$

$\Delta_i = d_i(x) - d_i^{\text{Af}}$; $\sigma_{i,\text{min}}$ comes from the fitted distogram width.

A Gaussian distance likelihood is marginalized over $\sigma \geq \sigma_{\text{min}}$ with a $1/\sigma$ prior.

At $\Delta_i = 0$, use $L_i = 1/(\sqrt{(2\pi)} \sigma_{i,\text{min}})$.

bAles status for Tau-5*

WT and AA: all five AF2 models and $120 \times 120 \times 64$ histograms are stored locally.

Local LAMMPS and PLUMED compilation completed; the tutorial smoke test stops at a PLUMED API-version compatibility check.

No bAles Tau-5* production ensemble is available yet. AF2 model files are not a substitute for that ensemble.

DENSS and independent information

$$\langle |F(\mathbf{q})|^2 \rangle \neq |\langle F(\mathbf{q}) \rangle|^2$$

$F(\mathbf{q})$: scattering amplitude. An ensemble's intensity is not generally the intensity of its mean density.

DENSS reconstructed from the same SAXS curve is not a new independent experimental restraint.

Use density comparisons as structural interpretation with stated reconstruction/alignment assumptions; avoid counting the same SAXS information twice.

Testing cooperative helices and hydrophobic contacts

Vary helix-length and S α distributions while retaining comparable residue-wise secondary-structure populations.

Stratify candidate pools by hydrophobic SASA or contact coordination; select with the same CS–SAXS objective.

Compare fit quality and structural distributions across independently generated pools. Add independent observables to distinguish equally good fits.

What the current results establish

Equal-weight selections reach carbon RMSDs of roughly 0.29–0.43 ppm with 500 distinct structures.

WT SAXS remains less well fit than AA; the small helix-length supplement has only modest effects.

The next decisive tests are independent pools, stronger cooperative-helix sampling and convergence of contacts as well as local helicity.

Current analysis: code, selected members and numerical results

Key references 1 / 3

Zhu et al., Nature Communications (2022)

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Teixeira et al., J. Phys. Chem. A (2022)

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<https://doi.org/10.1038/s41467-026-69172-y>

bAles-IDP source and tutorial

<https://github.com/COSBlab/bAles-IDP>

Original AR analysis code and MD comparison

https://github.com/paulrobustelli/AR_ligand_binding

Current analysis: code, selected members and numerical results

https://github.com/paulrobustelli/tau5-sec-saxs-analysis/tree/main/Ensemble_Comparisons/CS_SAXS_Selection